Lung Function Estimation from a Monosyllabic Voice Segment Captured Using Smartphones

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ABSTRACT

Chronic respiratory diseases refer to a group of lung diseases that affect the airways and cause difficulty in breathing. Respiratory diseases are one of the leading causes of death and negatively impact the patients' quality of life. Early detection and regular monitoring of lung functions might reduce the risk of death; however, lung function assessment requires the active supervision of a medical professional in a clinical setting. To make lung function tests more accessible and ubiquitous, researchers started leveraging mobile devices, which still require active supervision and demand extraneous effort from the user. In this work, we propose a convenient mobile-based approach that uses a monosyllabic voice segment called 'A-vowel' sound or 'Aaaa...' sound to estimate lung function. We conducted two studies (a lab study and an in-clinic study) with 201 participants to develop a detection model detecting 'A-vowel' sound from other acoustic events and a prediction model to estimate the lung function using the detected A-vowel sound. Our study shows that A-vowel sounds can be detected with 93% accuracy, and A-vowel sounds can estimate lung functions with 7.4-11.35% mean absolute error. We also conducted a validation study with 10 participants in a noisy environment and able to detect A-vowel segments with 71% F1-Score. Our results show auspicious directions to expand the horizon of mobile-based lung assessment.

CCS CONCEPTS

• **Human-centered computing** → **Ubiquitous and mobile computing design and evaluation methods**; Empirical studies in ubiquitous and mobile computing.

KEYWORDS

A-vowel Test, Respiratory Assessment, Digital Health

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1 INTRODUCTION

Chronic respiratory diseases (CRDs), including chronic obstructive pulmonary disease (COPD) and asthma, are the third leading causes of deaths in the world, accounting for more than 3.9 million deaths worldwide [31]. CRDs impose a significant socio-economic burden on patient [21] and negatively affect the quality of life [9, 18, 46]. Early detection and regular monitoring of lung function can reduce the disease progression and risk of death [5, 12]; however, regular monitoring can be challenging due to the lack of human and financial resources. The traditional approaches (e.g., spirometry test [20]) for lung assessment requires active supervision from a trained medical professional and need to be diagnosed in a clinical

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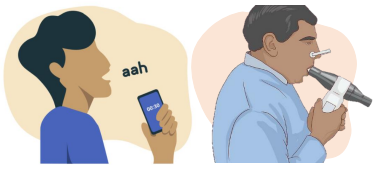


Figure 1: Subject holding the phone and speaking 'Aaaa...' vowel sound (left) and using a clinical Spirometer (right) [14] .

setting with sufficient resources. The quality of assessment also largely depends on the patient's cooperation and the experience of the technician performing the assessment [17].

To make lung assessment more ubiquitous, mobile-based lung assessment and monitoring have garnered significant attention from the research community [2, 7, 10, 16, 44]. However, the majority of prior works require extraneous effort from the patient, such as walking for six minute [3] or forced exhalation [16]. Such high effort activities can be challenging for severe and older patients as they may not be able to perform the activities, for example, people suffering from acute respiratory diseases might not perform forced exhalation [10]. Importantly, the accuracy of the existing methods depends a lot on the patients' effort, and a poorly performed task may invalidate the result [40]; therefore, mobile-based lung assessment still requires active supervision from a trained professional. Despite the significant effort from the researchers and clinicians, mobile-based lung assessment is still in the preliminary stage. To expand the horizon of mobile-based lung health interaction, we need to identify a more convenient and low-effort approach for lung assessment.

In this paper, we leverage a single segment of monosyllabic voice "Aaaa...", which we termed as 'A-vowel' sound, to estimate the lung condition. The ideation of using A-vowel is influenced by previous works [8, 30, 35, 41], where the results show that the voice quality of COPD and asthma patients during A-vowel sound could be affected by their diseases. A-vowel sound is a low-effort activity as it does not require forced exhalation, and it can be indicative of respiratory diseases. However, utilizing A-vowel sound to estimate lung function is still insufficiently explored. The feasibility of detecting A-vowel using smartphone audio and then use the A-vowel segments to estimate lung function is still unclear.

In this work, we present a low effort mobile-based approach to estimate the lung condition using A-vowel sound. To develop an A-vowel detection and lung assessment model, we conducted two data collection studies with 201 participants (including both Asthma and COPD patients as well as healthy control). The first study was conducted in a controlled environment (lab study) with 91 patients and 40 healthy participants. The second study was conducted in a clinical setting with 70 participants (10 healthy participants and 60 patients). In both studies, we instructed our participants to generate A-vowel sound as long as they can with multiple repetitions. We also collected the ground-truth of lung function estimation (FEV1/FVC ratio) using a commercially available spirometer (in the lab study) and clinical-grade spirometer (in-clinic study) to compare the A-vowel approach with gold-standard measurements. An example of our tests is depicted in Figure 1.

Our results show that we can estimate the lung functions with 7.4-11.6% mean absolute error using A-vowel sound collected from mobile devices. Since a user can give multiple 'A-vowel' sounds while performing the tests, our results also show that 'A-vowel' frames can be detectable from other sounds with 93.52% F1-score.

The main contribution of this paper are:

- We present a mobile-based approach to estimate lung functions using vowel sounds.
- Our work presents the development of a detection algorithm that detects A-vowel sound from other sounds.
- Our work also presents the development of a regression model that predicts the user's lung function using A-vowel sounds.
- Our work evaluates the detection and regression model using a robust dataset of 201 participants collected from different environments.

2 BACKGROUND AND RELATED WORKS

In this section, we begin by introducing background related to lung function measurements in general, and then we highlight some relevant works in the mobile-based lung assessment research domain.

2.1 Traditional Lung Function measurements

Spirometry is the standard test to measure lung function by estimating the amount of air the patient can inhale and exhale, and speed of exhalation. As shown in Figure 1, the patients need to use a spirometer tube into the mouth and attach a nose clip to perform the spirometry task. First, the patient needs to take a deep inhalation and exhale as hard as possible for several seconds into the tube. The patient needs to perform the test at least three times for a consistent result [6]. Based on the speed and volume of the airflow, the spirometer computes several metrics associated with lung functions, including the following three most widely used lung function measures:

- *Forced Vital Capacity (FVC):* This is the maximum amount of air an individual can exhale forcefully after a deep inhalation.
- *Forced Expiratory Volume in one second (FEV1):* This is the amount of air that an individual can exhale quickly and forcefully during the first one second of the forced exhalation.
- *FEV1/FVC ratio:* This is the ratio of FEV1 and FVC values, which is indicates an individual's lung function and help identify patient's lung obstruction.

These three lung function parameters are the most common clinically-reported measures to quantify the degree of airflow limitation in chronic lung diseases such as asthma and COPD. According to Global Initiative for Chronic Obstructive Lung Disease (GOLD), the FEV1/FVC ratio usually ranges between 0.7 and 0.8, and the ratio below 0.7 may indicate the presence of lung obstruction[1].

Challenges in using a spirometer: The risks of pulmonary function test may trigger dizziness, shortness of breath, coughing, and asthma attack due to forced breathing and effort involved in the test [19]. In some cases, a person should not perform a spirometry

[1]Gold Spirometry Guide. https://goldcopd.org/gold-spirometry-guide/

test because of issues including - (a) recent eye surgery, because of increased pressure inside the eyes during the procedure, (b) recent belly or chest surgery, (c) chest pain, recent heart attack, or an unstable heart condition, (d) a bulging blood vessel (aneurysm) in the chest, belly, or brain, and (e) active tuberculosis (TB) or respiratory infection, such as a cold or the flu [19]. Even when an individual can perform the test, many factors can make the measurement unreliable, such as patient effort, pregnancy, tiredness, or conditions that affect a person's ability to blow the air out of their lung.

2.2 Mobile Audio Based Lung Assessment

The majority of the works in audio-based respiratory assessment considered blowing exhalation sound captured in a smartphone microphone. For example, SpiroSmart [16] shows the feasibility of performing a spirometry test using audio data from a smartphone microphone. However, it has several challenges, including unsupervised usability issues, as it needs vigorous effort from the patients that can significantly elevate heart rate from the baseline. A patient with severely low lung function might not produce any blowing sound at all. To address some of the poor quality audio issues SpiroCall [10] uses an additional accessory with the smartphone called vortex whistler. On the other hand, SpiroConfidence [40] and ExhaleSense [25] attempted to categorize valid spirometry test from invalid efforts using machine learning algorithms on blowing sound data captured using a smartphone microphone. Smartphone spirometry is highly susceptible to poorly performed efforts because it needs specialized training with a sophisticated interface for the users to make sure they perform the test correctly.

In addition to mobile spirometry efforts, some other notable works considered passive audio data to assess lung functions. For example, Nemati et al., [23] and Ramesh et al., [28] used coughs to assess lung functions. Chun et al., [4] used passive speech audio to estimate the FEV1/FVC ratio. Although passive audio data collection involves low or no effort, they can pose privacy concerns to the users and the bystanders [13, 38, 39]. Our approach uses active audio recording; therefore, users are aware of possible audio recordings to adjust their privacy expectations.

2.3 Vowel Sound for Respiratory Assessment

Typically, vowel sounds contain a mixture of high and low-frequency overtones, and due to the filtering effect of air-filled lungs, vowel sounds are incomprehensible in healthy individuals. However, with CRD patients, the air in the lungs is replaced by fluid or solid substances, and for that reason, voice sounds can be better transmitted and well-distinct [34]. Due to this distinct voice sounds, vowel sounds have been used to diagnose CRD patients, and the process is called 'Egophony' [33]. Traditionally patients are asked to say 'Ee' in egophony which transformed to 'A' for patients suffering from respiratory diseases [33, 36]. However, the usage of other characters are also common, and various previous works [8, 35, 41] utilized different types of vocal sounds to diagnose COPD and Asthma patients. Dogan et al., [8] used 's' and 'z' phonate and showed that the maximum phonation time is significantly shorter for asthma patients. Shastri et al., [35] used 'Aaaa...' vowel and performed voice analysis on the data collected from 64 participants (32 COPD patients and 32 healthy participants). Their result also showed a significant difference between healthy participants and COPD patients on various acoustic measures, including lower fundamental frequency, reduced frequency range, increased pitch, amplitude perturbation measures, and increased noise measures. Walia and Sharma [41] also used 'Aaaa...' vowel sound and proposed a numerical formula to determine the levels of asthma. Considering the usage of 'Aaaa...' in previous works, we also used 'A-vowel' sound to assess the lung function of CRD patients. However, none of the previous works used A-vowel sounds to estimate the lung obstruction. The feasibility of using A-vowel sound on mobile devices using mobile microphones to assess lung function has not been explored yet.

	Lab Study	In-Clinic Study	Validation Study
Total Participants	131	70	10
Healthy Participants	40	10	8
Participants with CRDs	91	60	2
Age range	18-68	21-80	25-70
Gender	67 M, 64 F	28 M, 42 F	2 M, 8 F
Compensation	$150	$100	$100

Table 1: Summary of study participants. (M: Male, F:Female)

3 DATA COLLECTION STUDIES

We conducted two studies with 201 participants to collect 'A-vowel' ('Aaa...') audio data from healthy participants and patients with respiratory diseases. The first study was conducted in a controlled lab environment with 131 participants (*Lab Study*), and the second study was conducted with 70 participants in a hospital setting (*In-Clinic Study*). We also conducted an additional study with ten participants to evaluate our proposed model (*Validation Study*). In all three studies, we asked our participants to perform various tasks, including voluntary coughs, A-vowel, spirometry, and speech; however, we only highlight the details of the A-vowel and spirometry sessions for the scope of this paper. A summary of our participants from three studies is presented in Table 1. The details of the study protocol are given below:

3.1 Study-I: Lab Study

3.1.1 Protocol. We used a Samsung Note 8 smartphone to collect audio data from mobile devices. We collected the ground-truth lung function parameters, such as FEV1, FVC, and FEV1/FVC ratio using an FDA[2] approved portable spirometer (GoSpiro[3]). During the study, the participants performed a series of tasks, including two spirometry sessions and one A-vowel session. During the spirometry task, we asked our participants to attach a nose clip and exhale on the spirometer as hard as possible. The spirometer was connected to a smartphone app, and the smartphone app helped us to assess the quality of the effort done by the participant. If the effort was not good enough, we asked the participants to perform the test again with more considerable effort. In the spirometry session, the participants performed the spirometry task three times. The best effort measured by the maximum FEV1/FVC ratio on the spirometer from each participant was considered as the ground-truths. In the A-vowel session, the participants were instructed to produce 'Aaaa...' as long as they can. We had sufficient gaps in between the spirometry sessions and the A-vowel session, including two one-minute silent breathing sessions, to provide resting time to the participants.

[2]Food and Drug Administration
[3]https://monitoredrx.com/

The whole study was audio-recorded using the smartphone, and we asked our participants to keep the smartphone nearby (within arm's reach) to get a good audio recording. More details of this study protocol and annotation methodology are available in our prior work [26].

3.1.2 Participants. As reported in Table 1, we collected data from 131 subjects, including 69 asthma patients, 9 COPD patients, and 13 with both conditions.

3.1.3 Ethical and Safety Considerations. Our lab study was approved by our Institutional Review Board (IRB). We also took necessary safety measures for conducting this study. We asked our participants to keep their rescue inhalers with them. We also excluded those whose conditions are severe to perform the tasks.

3.2 Study-II: Clinical Study

We partnered with Brigham Women's Hospital (BWH) to collect data in a clinical setting. Since this study was conducted in a hospital, there were some background noises, such as door opening and closing, people talking, and coughing. Data collected in this study had a significantly higher noise level (noise: -10.6 dB) compared to the controlled lab study (noise: -28.8 dB).

3.2.1 Protocol. In this study, we followed the same protocol as Study-I (Section 3.1.1) except that we used a hospital-grade spirometer[4] for collecting the ground-truth data. Similar to the lab study, the participants performed the spirometry and A-vowel task with sufficient gaps in between them. We only considered the best effort for our ground-truth. A certified respiratory technician was present during the whole study, helping the participant perform the task.

3.2.2 Participants. This study consisted of a total of 70 participants, including 25 asthma patients, 25 COPD patients, ten chronic cough patients, and ten healthy controls.

3.2.3 Ethical Consideration. Our study was approved by both our Institutional Review Board (IRB) and the hospital's review board.

3.3 Study-III: Validation Study

We use the data collected from the first two studies to develop an A-vowel segment detection model. We evaluated our model on the data collected from an independent study with ten participants (Study-III). In the study, participants produced speech, cough, and A-vowel sounds multiple times with deteriorating lung conditions. This study serves as an evaluation of our model's robustness as the performance was tested on varying lung conditions.

3.4 Data annotation and preparation

We recorded the start and end of each task through a companion app. We later listened to the audio recordings and annotated the start and end of each A-vowel segment. We captured 502 A-vowel segments from Study-I, 182 segments from Study-II, and 77 segments from Study-III. We annotated the onset and offset of each A-vowel segment. Since we have higher background noise in Study-II, two researchers separately labeled those A-vowel audio segments. We resolved the discrepancies in annotation by jointly viewing and discussing the segments. On average, the duration of an A-vowel segment was 8.12 seconds.

[4]Pneumotrac Portable Screening Spirometer. http://www.morgansci.com/pft-test-equipment-software/pneumotrac/

4 MODEL DEVELOPMENT

To assess the lung functions from the A-vowel task, we need the onset of A-vowel in the audio segment. Therefore, we need an A-vowel detection model to find the start and end time of the A-vowel segment. We then need a regression model to predict the FEV1/FVC values from the A-vowel segment. In this section, we discuss our methods of developing the A-vowel detection and the regression model.

4.1 Characteristics of A-vowel Sounds

For the A-vowel tasks, we observed notable differences in the raw audio signal of healthy participants and pulmonary patients. The raw audio signal and the envelope of the signal is shown in Figure 2 and 3. As shown in Figure 2 and Figure 3, the volume and duration of A-vowel tasks are much higher for the participants with higher FEV1/FVC. Additionally, the intensity of the sound gradually decreases for the participants with higher FEV1/FVC compared to the participants with a low FEV1/FVC ratio (Figure 3).

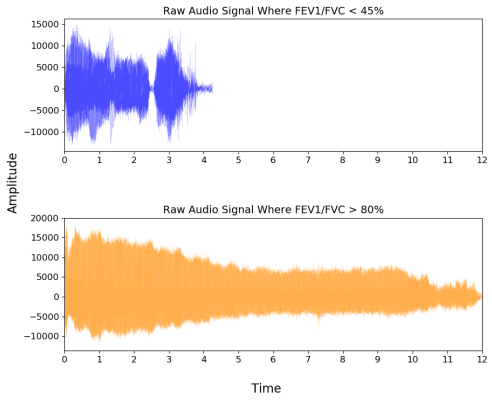


Figure 2: Waveform of A-vowel segment from two participants

4.2 System Overview

Figure 4 presents an overview of the data processing steps that we develop to estimate lung function. Audio signal processing starts with capturing the 'Aaa...' sound with user's smartphone microphone. As a user may produce several A-vowel sounds during one test session, we first need to detect and pinpoint the A-vowel segments in the audio stream. Therefore, after collecting the raw audio, the data is sent to the segmentation layer, where each A-vowel segment is detected from the captured audio stream. In the feature engineering layer, we compute a feature vector from each A-vowel segment. We then select a list of important features to train a regression model for estimating lung function in model development layer.

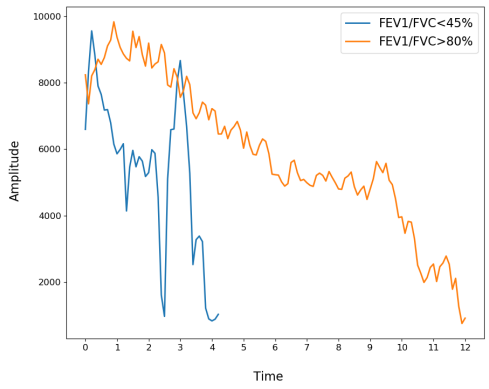


Figure 3: Envelope of the A-vowel segment from two participants

4.3 A-Vowel Detection Model

During an A-vowel task, the users are supposed to produce 'Aaaa...' sound as long as they can. In our experiment, the A-vowel segment can last one minute with multiple segments of A-vowel sounds. To extract important features from the voice segments, detecting the A-vowel segments from the one-minute voice segment is a prerequisite. We consider the detection task as a binary classification problem, where our goal is to detect A-vowel segments (*AV*) from other sound segments (*non-AV*). For the positive *AV* class, we use A-vowel segments, and for the negative *non-AV* class, we use silence, cough, speech, forced exhalation, and noise segments. In this section, we discuss our classification steps to detect A-vowel segments:

4.3.1 Pre-processing. First, we filter the audio stream using an exponentially weighted moving average filter and scale the resulting data to the unit norm (L_2 normalization). To remove the background noise, we perform loudness normalization, which eliminates bias due to the variation of perceived loudness across different sound frames in each audio window [27]. These procedures remove most of the background noises.

4.3.2 Segmentation. After removing noise from the audio signal, We extracted *frames* from the pre-processed data streams using a traditional window-based approach with a 50% overlap. The frame size plays a vital role in classification. We segment the whole time-series into 200 ms frames.

4.3.3 Feature Extraction. We computed five statistical functions for each frame: the audio signal's mean, variance, skewness, kurtosis, and power. These frame-level features comprise a commonly used representation for the underlying audio data. For an audio signal with n ms length, the end result of the feature extraction step were $(\frac{n}{200}\mathbf{x}5)$-dimensional feature vectors.

4.3.4 Dataset. We have 37,950 A-vowel frames from the lab study data and 11,112 A-vowel frames from the in-clinic study data.

4.3.5 Model Evaluation. We trained three models: 1) *Model-1 (Lab Study)*: we used two-thirds of the lab study to train the model, and the rest of the lab study data was used for testing; 2) *Model-2 (Cross Study)*: we used the lab study data to train the model and in-clinic data to test, and 3) *Model-3 (In-Clinic Study):* we used two-thirds of the in-clinic data to train the model and the rest of in-clinic study data was used for testing.

Table 2: Train and Test set size for each model

	Model-1 (Lab Study)	**Model-2** (Cross Study)	**Model-3** (In-Clinic Study)
Train	24,021	37,950	7,449
Test	13,929	11,112	3,663

4.3.6 Learning Algorithm. We use decision tree (DT), Naïve Bayes (NB), linear regression (LR), Support Vector Machine (SVM), Random Forest (RF) with 100 trees, and Ada-Boost (AB) classifier to classify the 200ms audio frames. We use grid-search to find the best hyper-parameter of the model. To evaluate the performance of learning algorithms, we evaluate the performance of the classification accuracy for each 200 ms frame. We use recall, precision, and F1-score as evaluation metrics.

4.3.7 A-Vowel Segment Construction. As there can be several clusters of *AV* frames after the classification, we need some post-processing to identify the start and end of an A-vowel segment. We use a simple clustering algorithm based on a time difference threshold for grouping classified A-vowel frames to form a cluster. When two consecutive inferred A-vowel frames are within a specific temporal distance, we put these two frames in the same cluster. After clustering, we use a simple rule-based method to declare a cluster of A-vowel frames as the A-vowel segment based on minimum duration. We consider the time of the first frame as the start time of and time of the last frame as the end time of each A-vowel segment. We only have a segmented audio frame from the lab and in-clinic study. To evaluate the post-processing algorithm's performance, we run our post-processing algorithm on the validation study dataset.

4.4 Lung Function Estimation

After detecting the A-vowel segment, we first extract additional features from the audio segments and then conduct feature selection and transformation. In this section, we discuss the feature extraction, feature selection, and regression model development for estimating the lung function.

4.4.1 Feature Extraction. We have extracted a set of 310 features for estimating lung function using regression models. In this section, we discuss the features:

Envelope Features. The time-domain envelope is taken using the Hilbert envelope. The Hilbert transform of the signal is then passed through a low pass filter to extract the envelope. We use the area under the envelope (*env_area*) and *duration* of the A-vowel segment in our list of features.

Time and Frequency Domain Features. We divide each A-vowel segment into frames of 20ms with a 50% overlap using a sliding window-based approach. We compute two energy features (*energy* and *energy_entropy*) and Zero Crossing Rate (*zcr*) from from each audio frame. We also extract 31 frequency level features from each

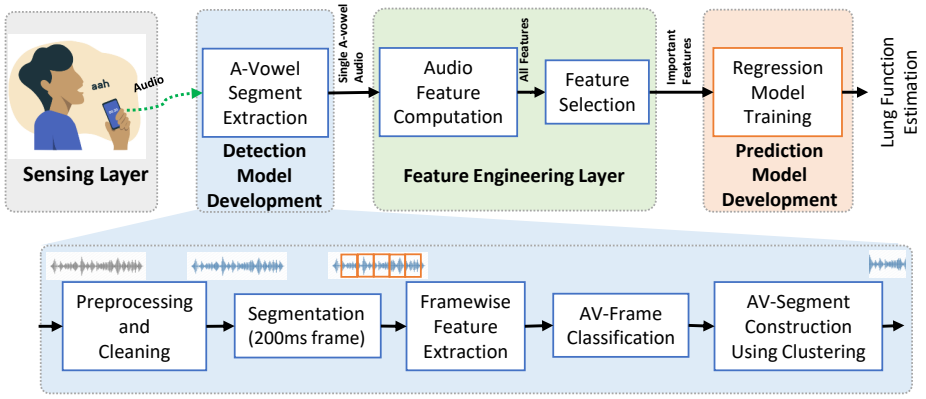


Figure 4: Overview of key modeling steps for estimating lung function

audio frame, including Spectral Centroid, Spectral Spread, Spectral Entropy, Spectral Flux, Spectral Rolloff, 13 Mel Frequency Cepstral Coefficients (MFCC), 12 Chroma, Standard Deviation of 12 chroma features. In total, we have 34 different types of time and frequency domain features. We applied four aggregated statistical functions (mean, standard deviation, kurtosis, and skewness) on the time and frequency domain features. Therefore, we extract 34×4=**136** features from the audio frame.

Temporal Features. As can be seen in Figure 2, during an A-vowel task, the intensity of the signal decreases over time. While the beginning part of the A-vowel audio tends to have high energy, the end part tends to have low energy. To capture this phenomenon, we divide the whole A-vowel segment into two equal parts: *left* and *right*. We compute the average and standard deviation of the time-domain and frequency-domain features for both parts of the audio segment. Additionally, we also compute the differences from left value to the right value for each time and frequency domain features, which capture the changes of the last part of the signal from the beginning part. Therefore, we extract 5×34 =**170** temporal features from each audio frame.

Speech Features. In addition to the 308 features, we use two speech features: shimmer and jitter to assess lung function [22, 32]. *Shimmer* refers to the consistency of localized vocal amplitude for vowel sounds, and *jitter* refers to the consistency of localized periodicity of the same.

4.4.2 Feature Selection. We conduct feature selection and transformation to overcome problems related to multi-collinearity and covariance that can potentially affect downstream regression tasks. To reduce feature space, we applied principal component analysis (PCA) [15] using a singular value decomposition solver [42] on the features mentioned in Section 4.4.1. This method reduces the dimensions in the feature space by transforming features into orthogonal or principal components [15, 45]. The explained variance based on the number of principal components is shown in Figure 5 indicates that 20 feature approximately contains 80% of the variance on both data set. For our regression task, the size of the final feature set is 20. Since the PCA method does not give the selected feature names, we use mutual information to observe the most important features [24].

If the mutual information between the target FEV1/FVC and the feature mutual information is greater than 0.5, we consider that feature as an important feature. The top ten features for FEV1/FVC based on mutual information are: *spectral_entropy_std, energy_mean, energey_mean_left, energy_mean_right, env_area, energy_centroid_std, zcr_std, zcr_std_right, zcr_std_left,* and *zcr_skew.*

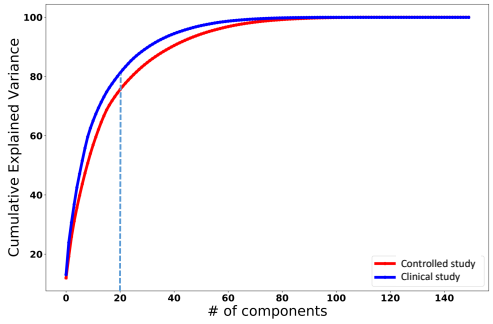


Figure 5: Plot of explained variance and number of PCA components to transform features. This plot help us to determine the number of components. 80% of the feature space explained by 20 principal components in the feature space.

4.4.3 Regression Model. We evaluated six regression models to estimate the FEV1/FVC ratio: Linear Regression, Linear Regression (LR-L1/L2) with both L1 and L2 normalization, Random Forest Regression (RFR), Bayesian Regression (BR), Gradient Boosting regression (GBR), Adaboost Regression (ABR), and Multilayer Perceptron Regression (MLR).

4.4.4 Evaluation. We use mean absolute error (MAE) for evaluating the performance of regression model. If $(y_1, y_2, ..., y_n)$ be the actual values of lung function and $(\hat{y_1}, \hat{y_2}, ..., \hat{y_n})$ be the predicted values then MAE is computed as,

$$MAE = \frac{\sum_{i=1}^{n} |y_i - \hat{y_i}|}{n}$$

5 RESULTS

In this section, we present the performance of our A-vowel detection and lung function estimation model:

5.1 Performance of A-vowel Detection

Table 3 shows the performance of each classifier for the detection task of A-Vowel segments. For all three models, precision is the highest for the Naive Bayes model, while the other measures are the lowest for this classification algorithm. Random forest algorithm yields the best results in terms of recall, F1-score, and accuracy. The F1-score of Model-2 (93.52%) indicates the robustness of the A-vowel detection model as it was trained on a different data set.

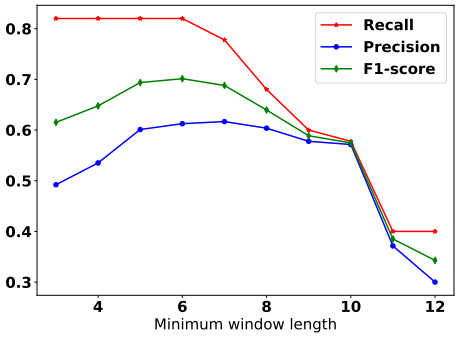


Figure 6: Performance of detecting A-Vowel segment with different window size (On the validation study data)

Detection of A-Vowel Segments. As discussed in Section 4.3.7, our approach for inferring the A-Vowel segment depends on the temporal density of observed 200 ms A-Vowel frames; we cluster these frames over time using the DBSCAN algorithm, which takes one parameter: minimum duration of a single A-Vowel segment. We used the data from our validation data set to measure the performance of detecting A-vowel episodes since we have segmented tasks from the first two studies. In the validation study, we have thirty minutes of voice data consisting of A-vowel, speech, cough, and noise. To assess how well A-Vowel segments were recognized, we compared the ground truth and predictions over different minimum duration lengths. Figure 6 plots recall, precision, and F1-scores as a function of minimum segment length ranging from 3 to 12 seconds. The charts show an upward trend of F1-score in recognition performance as segment length increases at the beginning, and then it starts going downward. When the segment length is set to 6 seconds: the F1-score is 0.71, recall is 0.85, and precision is 0.61. The performance of the classification algorithm can be considered pretty well, considering that the data set resembles real-world voice streams consisting of various sounds that were not available in our data set.

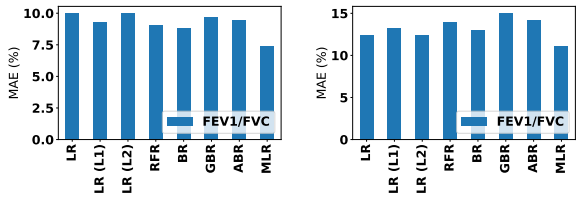
Boundary Error. We analyze the temporal alignment of the detected segment with the actual segment. We find that the error in alignment is small for both locating the onset and the end. The error in accurately locating the start and end times are 4.8% and 9.7%. The total duration error is 6.9%.

5.2 Performance of Patient Classification

Table 4 shows the patient classification results from leave-one-subject-out-cross validation. For both datasets, the model performs reasonably well with approximately 80% F1-score. For the clinical study, recall and precision are 89.47% and 73.91%, respectively.

5.3 Performance of Estimating Lung Function

Results of the regression model are shown in Figure 7. Multi-Layer Regression (MLR) model gives the best performance in estimating the FEV1/FVC ratio with a Mean Absolute Error (MAE) of 7.4% for the controlled lab study and 11.35% for the clinical study. Table 5 shows the performance of estimating lung function by different groups, such as asthma, COPD, and health. We observed that the mean absolute error for healthy participants is lower than other patients. The MAE of estimating the FEV1/FVC ratio of healthy participants is only 6.38% for controlled study and 5.15% for both the clinical study. This result indicates that the accuracy can be further improved if the patient's condition is known beforehand.



(a) FEV1/FVC ratio estimation (Lab study) **(b) FEV1/FVC ratio estimation (In-Clinic study)**

Figure 7: Mean absolute error between our prediction model and clinical spirometer for both datasets. LR: Linear Regression, RFR: Random Forest Regression, BR: Bayesian Regression, GBR: Gradient Boosting Regression, ABR: Ada-Boost Regression, and MLR: Multi-Layer Regression.

5.4 Effort Comparison

We hypothesize that an A-vowel test needs less effort from the user compared to a spirometry test. To test this hypothesis, we consider heart rate elevation to be an objective measure for the physical effort associated with the user's assessment activities and compare the heart rate between the A-vowel test and mobile spirometry [25]. Figure 8 shows the heart rate comparison between the A-vowel test, the mobile spirometry test, and the baseline resting heart rate. The figure shows that the heart rate during the A-vowel test lies in between the resting heart rate and the heart rate during the spirometry test. Although the A-vowel test causes heart rate elevation from the resting heart rate, we observe that the A-vowel test causes a significantly lower heart rate ($\mu = 77.12$) elevation compared to the mobile spirometry test ($\mu = 79.97$) ($p < 0.1$, paired t-test). Therefore, we expect that the A-vowel test will be more comfortable and easier to perform, especially for severe patients.

Table 3: Summary of classification performance (%) on detecting A-Vowel frames (200ms frames).

Classifier	Model-1 (Lab Study)				Model-2 (Cross Study)				Model-3 (In-Clinic Study)			
	Recall	Precision	F1	Accuracy	Recall	Precision	F1	Accuracy	Recall	Precision	F1	Accuracy
Decision Tree	91.08	94.98	92.99	90.05	80.89	92.30	86.22	81.03	90.04	91.58	90.80	86.63
Naïve Bayes	38.85	**97.92**	55.64	55.12	14.64	**98.76**	25.50	37.26	66.85	**98.59**	79.67	75.00
Linear Regression	97.03	96.50	96.76	95.30	81.82	97.37	88.93	85.05	84.49	95.53	89.67	85.74
SVM	97.10	96.66	96.88	95.48	82.32	98.09	89.52	85.86	89.95	94.55	92.19	88.83
Random Forest	**97.99**	96.66	**97.32**	**96.10**	**89.60**	97.80	**93.52**	**90.90**	**93.91**	92.77	**93.33**	**90.17**
Adaboost	95.56	95.90	95.74	93.83	85.47	95.15	90.05	86.15	90.35	95.03	92.63	89.47

Table 4: Leave-one-subject-out cross validation results for patient classification

Dataset	Recall	Precision	F1-score
Controlled study	83.72	75.00	79.12
Clinical study	89.47	73.91	80.95

Table 5: Lung function estimation based on participant's condition for lab and in-clinic study.

Participant Type	MAE(%)	
	(Lab Study)	(In-Clinic Study)
Asthma	7.20	10.72
COPD	10.05	11.09
Asthma + COPD	12.60	NA
Healthy	6.38	5.15

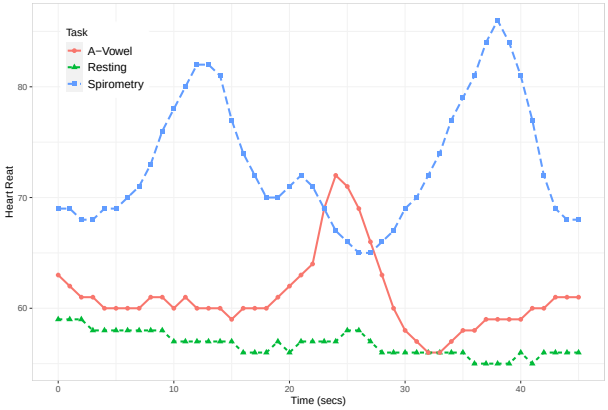


Figure 8: Physical exertion (measured in heart rate) of one participant while performing the tasks in our study: 1) when the subject was sitting at rest (green), 2) when the subject was performing A-vowel test (red), and 3) while performing two spirometry tests (blue). It shows that heart rate elevation during A-vowel is significantly lower compared to the mobile spirometry.

6 DISCUSSION, LIMITATIONS AND FUTURE WORKS

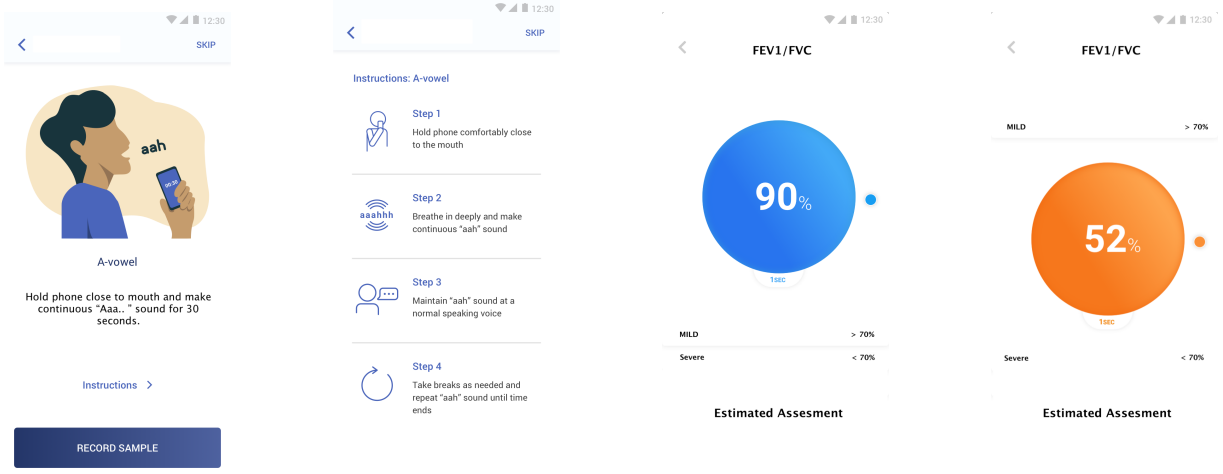
Potential Applications for Infectious Respiratory Diseases. Our model is developed based on the data collected from chronic respiratory patients. However, there are commonalities in changes in signs and symptoms across chronic respiratory patients (e.g., asthma, COPD) and infectious respiratory patients (e.g., COVID-19). For example, breathlessness, which is a critical symptom, increases during asthma and COVID-19 deterioration [11]. We observe in our study that participants with severe lung conditions took multiple breaks while producing 'Aaaa..' vowel sound, which could be useful for assessing breathlessness for COVID-19 patients.

Frequent respiratory assessments are the key to contain COVID-19 spread. However, there is a lack of assessment tools with higher prevalence in low and middle-income countries around the world. Our low-cost approach can instigate mobile-based, non-invasive assessment research, and help make the assessment more ubiquitous.

Moreover, smartphone-based respiratory assessment, similar to the model presented in this paper, can empower patients to monitor the progression of their diseases when they are confirmed from the pharyngeal swab test [1]. Multiple assessments can easily be performed to further increase the accuracy and reliability of the assessment through clinically acceptable data aggregation models. For example, a widely accepted RT-PCR test for COVID-19 frequently produces false-negative outcomes [43]. Multiple tests are performed with the same subjects at different points in time to confirm the COVID-19 negative test outcomes [37]. Future studies that analyze 'Aaaa...' sound from COVID-19 patients can confirm the feasibility and explore the clinical protocol for detecting the viral infection and the recovery.

Comparison with Spirometry. Spirometry requires significant effort from the user. During our data collection procedure, even healthy participants needed several attempts to produce a valid spirometry effort. An individual who never performed a spirometry test may struggle to produce a legitimate effort at his first attempt without clinical supervision. Thus, mobile-based spirometry might not be a suitable choice for in-home, early diagnosis. Compared to spirometry, the A-vowel task requires much low effort and supposed to be more intuitive. A-vowel test may not require additional training materials for the users to perform, yet it estimates lung functions reasonably. We want to note that a single A-vowel assessment may need more time (15 seconds) than that of a single spirometry test (6 seconds); however, A-vowel assessments are more convenient. Since A-vowel assessment involves less physical exertion, it will be less risky to flare up the respiratory condition further.

Mobile-based A-vowel Application. While participating in our study, our participants with respiratory conditions mentioned that they are comfortable using mobile devices for their lung function assessment. Considering the interests, we envision a mobile application where the user will use an A-vowel test to assess their lung function quickly. After starting the app, the user is instructed to say 'Aaaa...' as long as they can with multiple repetitions within 30 seconds.



(a) A-vowel task start screen **(b) A-vowel task instructions** **(c) UI when FEV1/FVC >70%** **(d) UI when FEV1/FVC <70%**

Figure 9: A sample A-vowel application for lung assessment.

After the data acquisition, the app shows the predicted lung function with intuitive color. Figure 9 shows a sample user interface. Note that, we used a similar user interface for the test initiation (Figure 9a) and instruction (Figure 9b) in our study. Moreover, estimated lung function (e.g., FEV1/FVC ratio) can be presented with color-coding to indicated the severity of the condition (as shown in Figure 9c and Figure 9d).

Using other vowel sounds. For this study, we used A-vowel sound to estimate lung function. However, other vowel sounds such as 'Eeee...' or 'Oooo...'. can also be a good indicator to assess lung function; therefore, other vowel sounds can be explored to extend the current work. Although 'E-vowel' sound has some correlation with lung functions [34], we do not have a good understanding of other vowel sounds. In the future, exploring other vowel sounds to estimate lung function could be an exciting direction. Additionally, we can consider combining multiple vowel sounds to improve accuracy. For example, combining both A-vowel and E-vowel might result in better accuracy.

Passive Lung Function Estimation on Extracted Vowel Segments. In our natural settings, we often use vowel-like sounds, which could be passively detected. We can consider a Voice Activity Detection (VAD) algorithm [29] to trigger the voice activity and then detect vowel-like sounds. From the vowel-like sounds, it is possible to extract intuitive features and estimate lung functions. Such short burst might not be indicative of anything; however, a compilation of such short segments could potentially indicate underlying lung diseases. Currently, the user needs to use the application actively and perform the test for 30 seconds. An undiagnosed CRD patient might not be motivated to perform the test. For those users, we can collect short vowel segments from the speech in a passive manner and estimate the lung function. However, this can raise significant privacy issues both for the user and the bystanders [38]. We need additional exploration for such passive estimation of lung functions.

Limitation. This work has several limitations that open up numerous opportunities for future works. During model training, we only use the maximum duration of the multiple A-vowel sounds as maximum effort. That may not always be true, and we need a better method to incorporate all the results from multiple tries from the same individual. Our model uses A-vowel sound as the source of input. In the clinical study, we observe that data contains louder sounds such as noise from the rotation of the analog fan, background speech, or door closing or opening sound. A better noise removal technique that can handle various types of noise could improve the results.

As we have a relatively large number of participants with varying conditions, some participants might have performed poorly. Therefore, our model already covers patient condition variations. However, our studies were conducted under supervision in controlled settings. More user variations will likely be unveiled when the application will be deployed in free-living. Future studies can include more user variations to improve the algorithm to handle unforeseen circumstances.

7 CONCLUSION

In this paper, we present a low effort-based approach to estimate lung function using A-vowel ('Aaa...') sound. Our result shows that it is possible to detect A-vowel sound using smartphone microphone data, and it is feasible to estimate the lung function by extracting intuitive features from an A-vowel sound. We conducted two independent studies for collecting data from chronic respiratory patients to cross-validate our models. Our work expands the horizon of mobile interaction by introducing a new method for home-based lung assessment. A-vowel assessments have the potential for a convenient mechanism to assess lung function; however, it needs further explorations.

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